Fetal Movement Analysis → Fetal Behavioral Profiling

EARLIER APPROACH
Detect fetal movement

→

CURRENT RESEARCH DIRECTION
Detect → Characterize → Profile fetal movement behavior

→

LONG-TERM OBJECTIVE
Behavioral indicators for fetal well-being + intuitive mobile visualization

RAW DATA

↓

COUGH	FOUR-IMU	OXFORD

↓

DATASET LOADERS
↓
COMMON DATA FORMAT
↓
TIMING / CONTINUITY
↓
ARTIFACT HANDLING — CURRENTLY HERE
↓
PREPROCESSING
↓
MOVEMENT DETECTION
↓
MOVEMENT CHARACTERIZATION
↓
FETAL BEHAVIORAL PROFILE
↓
ML / ANALYSIS
↓
MOBILE APPLICATION
↓
MOVEMENT VISUALIZATION / VIRTUAL FETUS

The architecture is intentionally being built layer-by-layer so that ML is applied only after the raw signals are converted into reliable, artifact-aware movement information.

Progress Report — Data → Timing Layer

1. Dataset Investigation — ✓ DONE
→
2. Dataset Loader Architecture — ✓ DONE
→
3. Loader Validation — ✓ DONE
→
4. Sampling / Timing Audit — ✓ DONE
→
5. Continuous Segmentation — ✓ DONE
→
6. Real-Data Timing Validation — ✓ DONE*
→
7. Artifact Handling — CURRENT

DATASET VALIDATION

COUGH
Loaded + standardized + timing validation

39 recordings → 124 segments

PASS ✓

FOUR-IMU
4-sensor independent stream validation + segmentation

109 recordings → 436 independent streams → 15,476 segments

PASS ✓

OXFORD
IMU + BCG loading + timing / segmentation

16 records → 16 continuous segments

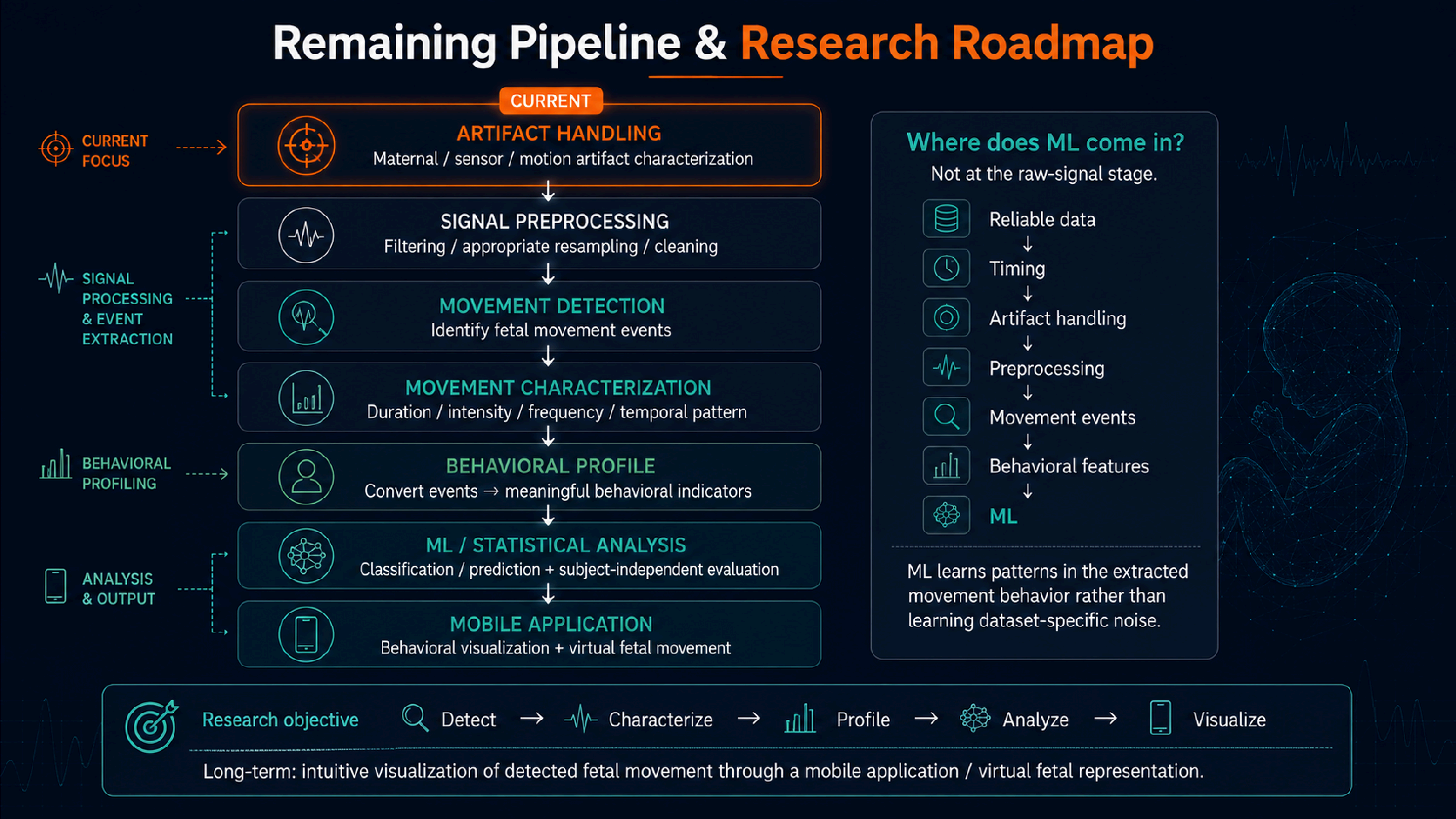
Validation correction ⚠

Implemented

- Dataset-specific loaders → common DataFrame
- Timestamp / sampling analysis
- Gap & discontinuity detection
- Continuous-segment generation
- Segment validation
- Metadata preservation
- Sensor identity preservation
- Real-data validation

✓ Real-data timing validation has now been performed across all three datasets.

Current focus → **Artifact Handling**


Remaining Pipeline & Research Roadmap
CURRENT FOCUS
SIGNAL PROCESSING & EVENT EXTRACTION
BEHAVIORAL PROFILING
ANALYSIS & OUTPUT
CURRENT
ARTIFACT HANDLING
Maternal / sensor / motion artifact characterization
SIGNAL PREPROCESSING
Filtering / appropriate resampling / cleaning
MOVEMENT DETECTION
Identify fetal movement events
MOVEMENT CHARACTERIZATION
Duration / intensity / frequency / temporal pattern
BEHAVIORAL PROFILE
Convert events → meaningful behavioral indicators
ML / STATISTICAL ANALYSIS
Classification / prediction + subject-independent evaluation
MOBILE APPLICATION
Behavioral visualization + virtual fetal movement
Where does ML come in?
Not at the raw-signal stage.
Reliable data
Timing
Artifact handling
Preprocessing
Movement events
Behavioral features
ML
ML learns patterns in the extracted movement behavior rather than learning dataset-specific noise.
Research objective
Detect → Characterize → Profile → Analyze → Visualize
Long-term: intuitive visualization of detected fetal movement through a mobile application / virtual fetal representation.